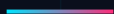


[Hugging Face](#) · [Stanford University](#)

SmolVLM: Redefining Small and Efficient Multimodal Models



Sub-1GB vision-language inference on edge devices

[arXiv:2504.05299](#) · [Fully Open-Source](#)

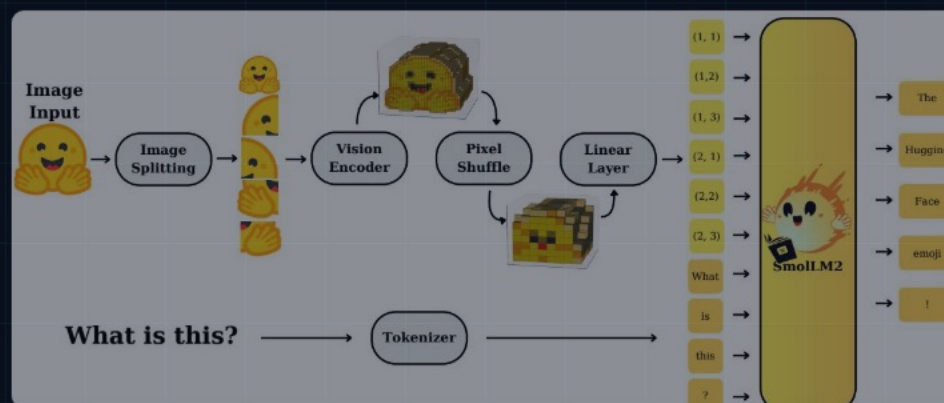


Large VLMs Are Powerful but Impractical for Edge Deployment

State-of-the-art vision-language models demand massive compute and memory, leaving mobile and edge devices underserved.



SmolVLM Architecture: Image Splitting, Pixel Shuffle, and SmolLM2 Backbone



Smaller Vision Encoders Outperform Larger Ones in Compact VLMs

Finding 1: Balanced encoder-LM compute allocation matters. A **93M SigLIP-B/16** paired with a 135M LM outperforms a **428M SigLIP-S0400M** with the same LM.

93M	428M
SigLIP-B/16	SigLIP-S0400M
Wins	Underperforms



Extending Context from 2K to 16K Tokens Unlocks Higher Image Resolution

Finding 2: Performance improves significantly with extended context. Achieved by increasing the **RoPE base from 10K to 273K** and fine-tuning on mixed long/short-context data.



SmolVLM Achieves Competitive Performance at Sub-1GB to 4.9GB RAM

Evaluated across 13 benchmarks spanning single-image, multi-task reasoning, and video comprehension.

Benchmark	SmolVLM 256M	SmolVLM 500M	SmolVLM 2.2B	Best Efficient OS
OCRBench	52.6%	61.0%	72.9%	54.7%
AI2D	46.4%	59.2%	70.0%	71.0%
ChartQA	55.6%	62.8%	68.7%	48.0%
TextVQA	50.2%	60.2%	73.0%	61.5%
DocVQA	58.3%	70.5%	80.0%	77.7%
ScienceQA	73.8%	80.0%	89.6%	87.5%
MMMU	29.0%	33.7%	42.0%	33.9%
MathVista	35.9%	40.1%	51.5%	37.6%
MMStar	34.6%	38.3%	46.0%	43.1%
Video-MME	33.7%	42.2%	52.1%	45.0%
MLVU	40.6%	47.3%	55.2%	48.2%
Average	44.0%	51.0%	59.8%	—

Real-Time Inference on iPhones: Up to 80 Tokens/Second on Apple Silicon

SmoIVLM runs entirely on-device via the [HuggingSnap iOS app](#) — no cloud dependency, full local privacy.

~80 tok/s · Pro M1 Max 64GB

~1000 tok/s · batch=64

~55 tok/s · MacBook Air M3



Key Takeaways: Design Principles for Efficient Multimodal Models

01 Balance encoder-LM compute. A 93M SigLIP-B/16 outperforms 428M encoders for small LMs.

03 Aggressive token compression. Pixel shuffle at $r=4$ yields 16 $\times$ reduction for small VLMs.

05 Fully open-source. Weights, datasets, code, and the HuggingSnap mobile app are public.

02 Extend context length. RoPE scaling to 273K enables 16K tokens, unlocking higher resolution.

04 Sub-1GB inference is achievable. 0.8 GB RAM at batch=1 on a 256M parameter model.

Implication: Efficient architecture > raw scale for on-device multimodal AI.

